

Homework 7

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Sep 19, 2025

Question 1:

(Grad version) Reproduce this plot as closely as you can. Hint: Read the plotmath documentation, and consider the construction `expression(paste(...))`.

```
# Set up the x-axis values
x <- seq(0, 8, length.out = 100)

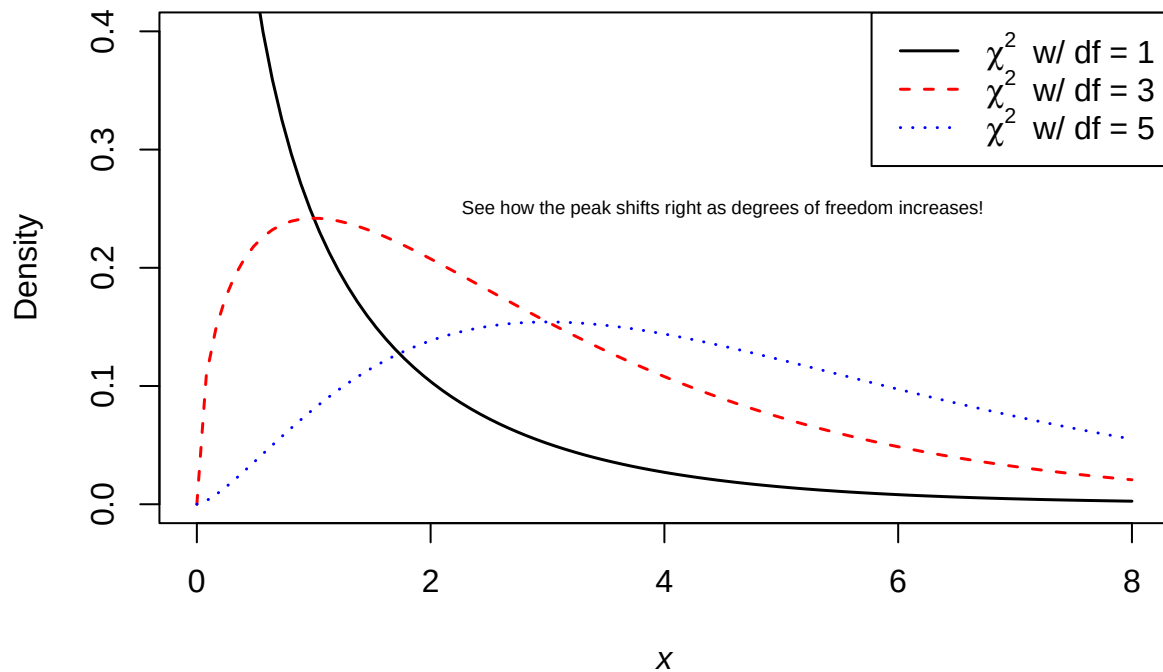
# Plot the first chi-square density curve
plot(x, dchisq(x, df = 1), type = "l", lwd = 1.5, col = "black",
      xlab = expression(italic(x)), ylab = "Density",
      main = expression("The " ~ chi^2 ~ " density"),
      ylim = c(0, 0.4))

# Add additional density curves
lines(x, dchisq(x, df = 3), col = "red", lwd = 1.5, lty = 2)
lines(x, dchisq(x, df = 5), col = "blue", lwd = 1.5, lty = 3)

# Add a legend
legend("topright", legend = c(expression(chi^2 ~ " w/ df = 1"),
                                expression(chi^2 ~ " w/ df = 3"),
                                expression(chi^2 ~ " w/ df = 5")),
      col = c("black", "red", "blue"), lty = c(1, 2, 3), lwd = 1.5)

# Add annotation
text(4.5, 0.25, "See how the peak shifts right as degrees of freedom increases!", cex = 0.6)
```

The χ^2 density



Question 2:

reproduce this plot as closely as you can. I'll get you started with the first two lines that produce the data. Create the appropriate formula objects to pass into `lm()`, and extract the `$fitted.values` object to plot the prediction curve. Hint: To avoid repetition in your code, can you use `lapply()` to iterate over degrees and colors?

```
# Generate data
x <- seq(1, 10, 0.5)
y <- sin(x)

# Scatter plot of original data
plot(x, y, col = "orange", pch = 1, ylim = c(-1, 1),
     main = "Polynomial approximation of sin(x)",
     xlab = "x", ylab = "y")

# Degrees and corresponding colors
degrees <- 1:5
colors <- c("red", "blue", "green", "orange", "gray")

# Fit models and add lines
fits <- lapply(degrees, function(d) {
  model <- lm(y ~ poly(x, d, raw = TRUE))
  lines(x, model$fitted.values, col = colors[d], lwd = 2)
```

```

return(model)
})

# Add legend
legend("top",
      legend = paste("Degree", degrees),
      col = colors, lwd = 2, bty = "n", cex = 0.8)

```

